

Learning to Spread Edges: Reinforcement-Learning Construction of Spatially-Coupled LDPC Codes

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Abstract—Spatially-coupled low-density parity-check (SC-LDPC) codes are built from a component protograph by *edge spreading*: each systematic base edge is assigned to one of $w+1$ coupling components $B_0, \dots, B_w$. In standard 5G-NR-based constructions this assignment is left to a random number generator, yet we measure that this single random choice changes the frame-error rate (FER) at a fixed operating point by roughly $7\times$. We make four contributions. (C1) We show that edge spreading is a high-leverage but routinely randomized design freedom: optimizing it alone lowers FER by $5\text{--}11\times$ and *eliminates the error floor*; the gain holds across code rates $0.5\text{--}0.9$, with a waterfall-threshold gain of $0.12\text{--}0.92$ dB that is largest (~ 0.9 dB) at $R \approx 0.6$ and shrinks monotonically toward high rate. (C2) We cast edge spreading as a Markov decision process (MDP) and optimize it end-to-end with policy gradient (REINFORCE), using the FER of the real encode/channel/window-decode chain as the reward. The learned 8-parameter feature policy removes the dominant population of 4-cycles (girth $4 \rightarrow 6$, $n_4 \approx 930 \rightarrow 0$); we prove an exact, bit-exact-validated combinatorial lemma characterizing the 4-cycle count under edge spreading— $n_4 = \sum_{\text{keys}} Z \binom{m}{2}$ over (check-block-row, column-pair, shift-difference) keys—which explains why naive constructions carry ≈ 930 4-cycles, why no simple balance rule removes them (the shift-differences are fixed by the 5G-NR circulants, not by the construction), and hence why a search-based construction is needed. We further outline how finite-length protograph EXIT (PEXIT) analysis explains the rate dependence of the waterfall gain via threshold saturation. (C3) We give a deployable criterion for *when* reinforcement learning helps: its sample-efficiency advantage over blind random search grows monotonically with per-evaluation cost and search-space size $(w+1)^E$ —RL wins 13/15 long-code cells (advantage growing with w), while under a cheap surrogate it merely ties. (C4) The feature policy transfers *zero-shot* to new lifting sizes and rates, a deployment property that random search and the cross-entropy method (CEM) cannot offer.

Index Terms—Spatially-coupled LDPC codes, edge spreading, code construction, reinforcement learning, policy gradient, 5G NR, error floor, threshold saturation, quasi-cyclic codes.

I. INTRODUCTION

Spatially-coupled LDPC (SC-LDPC) codes couple a chain of otherwise independent LDPC component codes along a spatial dimension; under belief-propagation (BP) decoding of a terminated chain, their threshold *saturates* to the maximum-a-posteriori (MAP) threshold of the underlying component ensemble [1], [2]. This makes SC-LDPC a strong candidate for low-latency, streaming, and 6G channel coding, and the 5G-NR quasi-cyclic (QC) base graphs of 3GPP TS 38.212 [3] provide a standardized, efficiently encodable starting point.

An SC-LDPC code is obtained from a component protograph by *edge spreading*: writing the component base matrix as $B = B_0 + B_1 + \dots + B_w$, each base edge is assigned to exactly one component B_a , and the variable position t connects through B_a to check position $t+a$ [4]. Different edge spreadings are known to affect *both* the iterative BP performance and the distance/cycle properties of the resulting code [4]. Yet in practice—and in the original version of the open-source pipeline used here—this assignment is simply drawn from a random number generator. The single line `rng.integers(0, w+1, size=E)` is, in effect, an unexamined design decision.

How much does that decision matter? On a fixed channel realization (i.e. using common random numbers so that only the construction changes), we measure the FER of 14 *random* edge spreadings at one operating point and observe a spread of 0.106 to 0.725 —about a $7\times$ swing in FER from this one random choice alone (Section VII-A). The construction space is enormous: $(w+1)^E$ with E systematic base edges, e.g. $3^{40} \approx 10^{19}$ for the deep-dive configuration and as large as 31^{121} for the wide-memory sweeps. Edge spreading is therefore a high-leverage but routinely randomized construction freedom.

Contributions. We position this work as a finite-length *coding-engineering* paper in which a learning method is the tool and the code is the subject. Concretely:

- **C1 (the hard result).** Optimizing edge spreading—and nothing else—lowers FER by $5\text{--}11\times$ and *eliminates the error floor*; the gain holds over rates $0.5\text{--}0.9$, with a waterfall-threshold gain of $0.12\text{--}0.92$ dB, largest at $R \approx 0.6$ and shrinking toward high rate (Sections VII-B, VII-E).
- **C2 (mechanism and theory).** We cast construction as an MDP and optimize it end-to-end by policy gradient; the learned 8-parameter policy removes the dominant population of 4-cycles. We prove an exact, bit-exact-validated combinatorial lemma characterizing the 4-cycle count under edge spreading, $n_4 = \sum_{\text{keys}} Z \binom{m}{2}$ over (check-block-row, column-pair, shift-difference) keys (Section V), which explains why naive constructions carry ≈ 930 4-cycles, why no simple balance rule removes them (the shift-differences are fixed by the 5G-NR circulants), and hence why search-based construction is needed; and we outline a finite-length PEXIT objective that explains the rate dependence of the gain via threshold saturation (Section IV).
- **C3 (a methodological criterion).** The sample-efficiency advantage of RL over random search/CEM grows mono-

tonically with per-evaluation cost and search-space size $(w+1)^E$: RL wins 13/15 long-code cells with the advantage growing in w , but ties under a cheap surrogate (Sections VII-F, VII-G, VII-H). This is a deployable “when to use RL” criterion.

- **C4 (a deployment property).** The feature policy transfers zero-shot to new lifting sizes and rates—something random search/CEM, which produce a code-specific assignment vector, cannot do (Section VII-I).

The remainder follows the structure: Section II reviews SC-LDPC and edge spreading; Section III casts construction as an MDP; Section IV develops the threshold (PEXIT) objective; Section V states the combinatorial lemma; Section VI describes baselines and setup; Section VII reports results; Section VIII discusses limitations, including a scoped negative result on the error floor; and Section IX concludes.

A. Related Work and Distinction

We organize prior art along two axes: decoding versus construction, and block versus spatially-coupled codes.

RL for LDPC decoding is mature: RELDEC casts cluster scheduling as an MDP and learns sequential schedules for 5G-NR LDPC [5]; our companion line learns sliding-window scheduling for SC-LDPC. In all of these the code is *given*.

AI/RL for code construction exists but not for spatial coupling. “AI Coding” [6] proposes a constructor–evaluator framework (RL or a genetic algorithm) for *linear block / polar* codes; a deep-RL/MCTS approach learns PEG-style edge growth for LDPC *block* codes [7]. Neither targets SC-LDPC nor edge spreading.

SC-LDPC construction has been studied with non-learning search: Mitchell–Lentmaier–Costello define edge spreading and quantify its effect [4]; high-rate array-based edge spreading is designed in [8]; greedy elimination of harmful objects is used in [9]; the partition matrix is optimized by exhaustive traversal with pruning in [10], [11], whose authors note that such discrete optimization “struggles as more degrees of freedom appear”; a 6G edge-spreading Raptor-like construction [12] and an MCMC sampler [13] are likewise non-RL. After an extensive literature check, *no published work casts SC-LDPC construction / edge spreading as an MDP and optimizes it with reinforcement learning*; existing methods are greedy, exhaustive, graph-theoretic, or MCMC. Our methodological template is the neural combinatorial optimization framework of Bello *et al.* [14] with the REINFORCE estimator [15]. Table I summarizes the distinction from the closest works.

II. BACKGROUND

A. From Block Codes to a Coupled Chain

An LDPC block code is defined by a fixed parity-check matrix H (Tanner graph). An SC-LDPC code couples a chain of LDPC component codes—each described by a protograph / base matrix B of size $m_b \times n_b$ —along the spatial dimension. First introduced as LDPC convolutional codes [2], the protograph view was systematized by [1].

TABLE I
DISTINCTION FROM THE CLOSEST WORKS.

	AI Coding [6]	Esfahanizadeh / Battagioni	This work
Code	block/polar	SC-LDPC	SC-LDPC
Object	gen./parity	partition matrix	edge-spread assign
Method	RL / GA	greedy/exhaustive/MCMC	policy-gradient MDP
Reward	dist./thr./BER	asy. thr. + cycles	end-to-end FER
Transfer	—	no (re-search)	yes (zero-shot)

B. Edge Spreading and the Band-Diagonal Matrix

Given coupling memory w , edge spreading splits B into $w+1$ component matrices,

$$B = B_0 + B_1 + \dots + B_w, \quad (1)$$

with each base edge assigned to exactly one component. Stagging the components along the diagonal yields a band-diagonal coupled parity-check matrix with block structure

$$H_{SC}[i, t] = B_{i-t} \quad \text{for } 0 \leq i - t \leq w, \quad (2)$$

where $t = 0, \dots, L-1$ indexes variable (column) positions and $i = 0, \dots, L+w-1$ indexes check (row) positions over a coupling length L . Variable position t thus connects through $B_0, \dots, B_w$ to check positions $t, \dots, t+w$.

C. Termination and Rate Loss

The w extra check positions at the tail are the *termination*: they impose stronger constraints at the chain boundaries (lower-degree, more reliable checks), at the cost of a small rate loss. The coupled rate R_L satisfies $R_L \rightarrow R$ as $L \rightarrow \infty$, with rate loss of order w/L .

D. Threshold Saturation

The central theoretical property of SC-LDPC is threshold saturation [1]: the BP threshold of a terminated SC-LDPC ensemble saturates to the MAP threshold of its component ensemble. Termination triggers a *decoding wave* that propagates reliability inward from the boundaries; this is the mechanism by which suboptimal iterative decoding is lifted, essentially for free, to near-MAP performance. Sliding-window decoding [16] exploits the locality of this wave: BP is run only inside a window of W positions, so latency and storage scale with W rather than L . We use $W = w+1$ throughout.

E. 5G-NR Systematic Edge Spreading

We build SC-LDPC from the 5G-NR QC-LDPC base graphs of 3GPP TS 38.212 [3] (BG1: 46×68 , $K_b=22$; BG2: 42×52 , $K_b=10$). To preserve the efficiently encodable double-diagonal / accumulator parity structure of 5G NR, we spread only the *systematic* base edges (the first K_b columns) across $B_0, \dots, B_w$ and keep the entire parity part in B_0 . Each position is then encoded by the standard 5G recursion, and the coupled $H_{SC}c = 0$ holds exactly. The set of E systematic base edges is the construction variable; the parity structure and

QC lifting shifts are fixed. Why edge spreading is a natural fit for sequential optimization: it is an inherently sequential assignment process (each edge in turn picks a component), its quality is governed by the cycle structure of the lifted graph, and that structure is highly sensitive to each edge's assignment—precisely where a policy gradient can climb a structural gradient while blind search cannot efficiently cover $(w+1)^E$.

III. CONSTRUCTION AS A MARKOV DECISION PROCESS

We construct a code edge-by-edge in a fixed row-major order $e_0, \dots, e_{E-1}$ and cast it as an MDP.

A. State, Action, Reward

- **Episode:** construct one full code (assign all E edges) and evaluate it end-to-end.
- **Time step t :** choose a component $a_t \in \{0, \dots, w\}$ for edge e_t .
- **State s_t :** features of edge e_t under the *current partial construction*. For each candidate component a we compute the fraction of edges in the same base *row* already assigned to a (`row_frac`), the same fraction by base *column* (`col_frac`), the *global* load fraction of a (`glob_frac`), and binary indicators of whether a is still empty for this row / column (`row_empty`, `col_empty`). These running counts make the process a genuinely sequential MDP and let the policy learn structural rules such as balancing component loads within a row or column.
- **Action a_t :** the component index.
- **Reward:** intermediate rewards are 0; at the end of the episode a terminal reward $R = -\text{FER}$ is given, obtained by placing the constructed code into the *real* encode $\rightarrow$ BPSK+AWGN $\rightarrow$ sliding-window BP $\rightarrow$ FER chain at the training SNR over a batch of common-random-number (CRN) frames. The ground truth is used only to score a constructed code at design time; the learned construction (the assignment vector) carries no ground truth and is directly deployable, exactly as for any code design.

B. Policy Gradient

Maximizing $\mathbb{E}[R]$ over the factorized policy $\pi_\theta(a) = \prod_t \pi(a_t | s_t)$ is REINFORCE [15], [14]:

$$\nabla J = \mathbb{E} \left[(R - b) \sum_t \nabla \log \pi(a_t | s_t) \right], \quad (3)$$

where b is an *in-batch baseline* (the mean of R over the batch). On top of this we use *common random numbers*: every candidate code in a batch is scored on the *same* frames, so the advantage $R_i - b$ has very low variance and the policy learns stably even when each code is scored on only a few frames. This variance reduction is what makes the method tractable on CPU. We optimize with Adam (pure NumPy), standardize advantages within a batch, and add an entropy bonus to encourage exploration. Both policy gradients are verified by finite differences.

C. Two Policies

- 1) **Feature policy (primary).** A linear softmax over the per-component features, with parameters $w+1$ component biases plus a handful of shared weights—8 *numbers in total* for $w=2$, *independent of the code size*. Hence the learned policy can transfer zero-shot to codes with different $Z/L/\text{rate}$ (provided w matches), and it is interpretable.
- 2) **Per-edge policy (ablation).** Independent logits per edge ($E \times C$ parameters): maximally expressive but non-transferable, used as a control for whether a generalizing structure is needed.

D. Algorithmic Notes

The implementation is pure NumPy with multiprocessing over CPU cores; the masks (puncturing/termination/transmitted positions) depend only on L, w, K_b, n_b, Z and *not* on the assignment, so all candidate codes can be scored on bit-identical information and noise (CRN), which the evaluator is tested to reproduce bit-exactly. The champion of each method is re-tested on an *independent*, larger frame bank to remove selection noise.

IV. A THRESHOLD (PEXIT) OBJECTIVE

This section develops a *frame-free, principled* objective and predictor that (a) addresses the “Monte-Carlo only” concern, (b) *explains* the rate dependence of the waterfall gain observed in C1, and (c) is the sharpest instance of the C3 “expensive evaluation $\Rightarrow$ RL wins” criterion.

A. Why Asymptotic Thresholds Are Nearly Construction-Invariant

By threshold saturation (Section II-D), the asymptotic BP threshold of a good terminated SC-LDPC ensemble approaches the component MAP threshold, *independently of fine construction detail*. This is precisely why the asymptotic threshold cannot, on its own, separate two good edge spreadings: the BP-MAP gap that coupling fills is large at low rate (strong component, many checks per position) and small at high rate (weak component). Consequently the *waterfall* construction gain is largest at moderate rate and shrinks toward high rate—exactly the trend measured in Section VII-E. What *does* separate finite-length constructions is the finite-length PEXIT recursion including the termination boundary, together with the cycle structure.

B. Protograph EXIT for the Coupled Graph

We adopt protograph EXIT (PEXIT) [17] over the band-diagonal protograph defined by the assignment (components $B_0, \dots, B_w$, coupling length L , and the termination boundary), using the BIAWGN J/J^{-1} mutual-information transfer functions and the finite-length protograph analysis of [18]. The recursion returns (i) the E_b/N_0 threshold required for convergence and (ii) a finite-length PEXIT BER proxy. As a reward this is plugged into the same pipeline via $R = -\text{threshold}_{E_b/N_0}$, reusing the REINFORCE machinery unchanged and replacing the Monte-Carlo evaluator with a PEXIT evaluator.

TABLE II
PEXIT VALIDATION (PLACEHOLDER—EXPERIMENT PENDING).

Quantity	Value
Spearman(ρ): PEXIT thr. vs MC waterfall thr.	[TODO: pending]
PEXIT-reward champion FER (MC re-test)	[TODO: pending]
MC-reward champion FER (MC re-test)	[TODO: pending]
PEXIT eval cost / MC eval cost	[TODO: pending]

C. Honest Scope

We do *not* claim that an asymptotic PEXIT threshold strongly discriminates constructions—threshold saturation makes the asymptotic thresholds of good constructions converge, which is itself the theory to be told. The discriminating quantity is the *finite-length* PEXIT recursion (with termination) plus the cycle structure. The deliverable of this section is therefore an *explanation* and a *principled predictor*, not an overwhelming new reward.

D. Validation Experiments (Pending)

The PEXIT objective is specified but the numerical validation is not yet run. Three experiments are planned:

- 1) Spearman correlation between the PEXIT threshold ordering and the Monte-Carlo waterfall-threshold ordering, to establish the proxy’s validity. [TODO: run PEXIT-vs-MC Spearman experiment; see TCOM_PLAN Sec.5(1).]
- 2) Train RL with the PEXIT reward, then re-test the champion by Monte-Carlo and compare against the MC-reward champion. [TODO: run PEXIT-reward RL training + MC re-test.]
- 3) Measure the PEXIT evaluation cost versus the Monte-Carlo cost to reinforce C3. [TODO: measure PEXIT vs MC evaluation cost.]

Numerical results are deferred; the placeholder table is Table II.

V. AN EXACT COMBINATORIAL CHARACTERIZATION OF THE 4-CYCLE COUNT UNDER EDGE SPREADING

We now characterize exactly how edge spreading controls the lifted-graph 4-cycle count n_4 . The result below is not an interpretable “spread evenly” rule—we show in fact that no such rule exists—but a closed-form count that makes precise *which* coincidences an optimizer must avoid, and thereby motivates the search-based construction of Section VII-D.

A. Setup and Notation

Fix the 5G-NR base graph with m_b systematic check base rows and circulant size Z . A systematic base edge e has base row r_i , base column c_j , and a QC circulant shift v that is *fixed* by the 5G-NR base graph, not chosen by the construction. Assigning e to component $\text{comp} \in \{0, \dots, w\}$ places it, for each spatial position t , as a coupled base entry in

$$\text{check-block-row CBR} = (t + \text{comp})m_b + r_i, \quad (4)$$

at variable-block-column $\text{VBC} = tn_b + c_j$, carrying the fixed shift v . Thus the construction (the assignment) controls

only which coupled check-block-row each base entry lands in, through the time-offset comp; it changes neither the column c_j nor the shift v .

B. Lemma Statement

Lemma (exact 4-cycle count). Consider the coupled, QC-lifted Tanner graph produced by an edge spreading. Within a single coupled check-block-row, associate to each unordered pair of coupled base entries, at variable-block-columns (C_a, C_b) with fixed QC shifts (v_a, v_b) , the key

$$\text{key} = (\text{CBR}, (C_a, C_b), d), \quad d = (v_a - v_b) \bmod Z. \quad (5)$$

By the standard QC 4-cycle condition [19], two entry-pairs sharing the same key close exactly Z lifted length-4 cycles. Hence if m base entry-pairs share a key, they contribute $Z \binom{m}{2}$ lifted 4-cycles, and the total 4-cycle count is

$$n_4 = \sum_{\text{keys}} Z \binom{m}{2}. \quad (6)$$

C. Proof Sketch

A length-4 cycle in a QC Tanner graph passes through two variable blocks and two check blocks connected by four circulant edges; by [19] the number of such lifted cycles is governed solely by the differences of the circulant shifts at the two participating column blocks within a shared check block. By (4), two coupled base entries co-occur in the same check-block-row exactly when their (time, component) offsets coincide, i.e. $t_a + \text{comp}_a = t_b + \text{comp}_b$ and they share base row r_i . Two entry-pairs in that check-block-row that further share the same column-pair (C_a, C_b) and the same shift-difference $d = (v_a - v_b) \bmod Z$ have, by the QC condition, exactly Z length-4 cycles between them. Summing $\binom{m}{2}$ over the m entry-pairs collected in each key, and Z per coincident pair, yields (6). Edge spreading enters only through *which* entries land in each check-block-row (the time-offset comp in (4)), hence through the per-key multiplicities m ; the column-pairs and shift-differences d are fixed by the 5G-NR circulants. Minimizing n_4 is therefore exactly minimizing $\sum_{\text{keys}} Z \binom{m}{2}$ by spreading entries so as to break up high-multiplicity (column-pair, shift-difference) coincidences within each check-block-row. This is the coupled, shift-aware analogue of the selective-cycle-avoidance principle of the ACE metric [20].

D. Why No Simple Balance Rule Suffices

Because the discriminating quantity is the (column-pair, shift-difference) coincidence and not same-column co-assignment per se, naive balance heuristics do *not* control n_4 . We verified this directly: the same-column, same-row, and global co-assignment counts each have a Spearman correlation with n_4 of $\rho \approx -0.02$, -0.13 , and $+0.12$ respectively (none predictive), and a deterministic max-column-spread construction yields $n_4 = 960$ —*worse* than round-robin (944), not zero. There is consequently no closed-form “spread-evenly”

rule that drives $n_4 \rightarrow 0$; an optimizer must reduce the high-multiplicity coincidences of (6) directly, which is precisely what RL, CEM, and a classical greedy/PEG/ACE local search all do (Section VII-D). This is the motivation for a search-based construction.

E. Numerical Validation

The closed form (6) is validated *bit-exact* against the repository’s `count_4cycles` routine: it reproduces the exact 4-cycle count on round-robin ($n_4 = 944$, girth 4) and on 30 random spreadings across all five deep-dive cells, with no discrepancy. Minimizing the actual `count_4cycles` via the classical baselines in `classical_baseline.py` (greedy harmful-object elimination, PEG, ACE) reaches $n_4 = 0$ and girth 6 on every cell; for reference round-robin gives $n_4 = 944$, the repository default (seed-0) 928, and random spreadings a median of ≈ 480 over the range $[0, 1424]$. The lemma is thus exactly confirmed and the optimized $930 \rightarrow 0$ reduction is reproduced by an exact combinatorial objective, independent of the end-to-end FER reward.

VI. BASELINES AND EXPERIMENTAL SETUP

A. Baselines

We compare the two RL policies against four baselines under identical evaluator, budget, and validation:

- **Random search** (key baseline): sample a batch of uniformly random assignments per step and keep the best by validation FER.
- **Cross-entropy method (CEM)**: a strong distribution-based optimizer maintaining per-edge categorical distributions and updating from elites.
- **Round-robin**: a naive perfectly-balanced heuristic.
- **Repository default (seed-0)**: the single random assignment shipped by the original pipeline.
- **Uncoupled component code**: the 5G-NR component code with no coupling.

All four optimizers (RL-feature, RL-per-edge, CEM, random search) share the same evaluator, the same evaluation budget (steps $\times$ batch), and the same validation protocol (fixed validation frames, then a high-precision re-test of the champion on a large independent frame bank). Any difference is attributable to the search strategy alone.

B. Configuration

The deep-dive operating point is BG2, $i_{LS}=0$, $Z=16$, $m_p=8$ (component rate ≈ 0.625 , coupled $R \approx 0.603$), $L=30$, $w=2$, sliding window $W=6$, normalized min-sum with $\alpha=0.8$, training SNR 2.5 dB, $E=40$. Each optimizer uses an equal budget of $32 \times 20 = 640$ construction evaluations at 36 CRN frames each; the champion is re-tested on an independent 2000-frame bank. Rate-sweep, long-code, and wide-memory configurations are detailed at the corresponding results subsections. Experiments were run on CPU (a workstation for the deep-dive, and 64–200 core cluster pods for the sweeps).

TABLE III
CONSTRUCTION DECIDES SUCCESS: 2000-FRAME RE-TEST AT 2.5 DB
(BG2, $Z=16$, $L=30$, $w=2$).

Construction	FER	BER	n_4	girth
CEM	0.064	1.37×10^{-4}	0	6
RL per-edge policy	0.080	1.98×10^{-4}	0	6
RL feature policy	0.092	2.67×10^{-4}	0	6
RL feature (greedy)	0.112	3.47×10^{-4}	0	6
Random search (budget 640)	0.133	2.60×10^{-4}	0	6
Round-robin	0.661	1.18×10^{-3}	944	4
Default seed-0	0.698	1.10×10^{-3}	928	4

TABLE IV
BER-VS- E_b/N_0 WATERFALL (DEEP-DIVE OPERATING POINT). “FLOOR”
MARKS THE HIGH-SNR SATURATION OF THE DEFAULT/NAIVE
CONSTRUCTIONS.

E_b/N_0 (dB)	2.2	2.5	2.8	3.1	3.4
RL feature	5.5e-3	3.4e-4	1.9e-5	5.1e-6	0
CEM	2.7e-3	1.4e-4	4.7e-6	0	0
Random search	3.9e-3	3.7e-4	3.7e-5	2.7e-6	0
Seed-0	6.2e-3	1.2e-3	3.5e-4	1.4e-4	6.4e-5 [†]
Round-robin	6.8e-3	1.1e-3	2.6e-4	1.0e-4	3.3e-5 [†]
Uncoupled	2.3e-2	9.2e-3	3.1e-3	1.0e-3	1.9e-4

[†] error floor.

VII. RESULTS

A. Headroom: Construction Alone Spans $7\times$

Using the CRN probe (same frames, only the construction changes) we measure the FER of 14 random edge spreadings at the deep-dive operating point: the FER spans 0.106 to 0.725, about a $7\times$ swing (BER spans $\approx 8.5\times$). This confirms (a) that the construction carries large headroom and (b) that 2.5 dB is an operating point with clear, non-saturated ordering—a good training point.

B. Construction Decides Success or Failure

Table III reports the high-precision 2000-frame re-test at 2.5 dB (lower FER is better), together with lifted-graph 4-cycle count n_4 and girth. Optimizing the single edge-spreading knob lowers FER from 0.70 to 0.06–0.13—about $5\text{--}11\times$. The repository default (seed-0) is among the *worst* constructions (0.698), and so is the naive perfectly-balanced round-robin heuristic (0.66).

The more important effect is on the *error floor*. Table IV and Fig. 1 give the BER-vs- E_b/N_0 waterfall: optimized constructions have *no* error floor (decaying to $\sim 10^{-6}/0$ at high SNR), whereas the default/naive constructions exhibit a floor at $\approx 3\text{--}6 \times 10^{-5}$. Measured at the BER = 10^{-3} waterfall threshold, the optimized constructions sit at 2.30–2.38 dB, the default/naive at 2.53–2.54 dB (~ 0.16 dB construction gain), and the uncoupled component code at 3.11 dB (~ 0.75 dB coupling gain). The construction dividend is therefore largest in the low-BER / error-floor region, consistent with the theory that short cycles mainly hurt the floor.

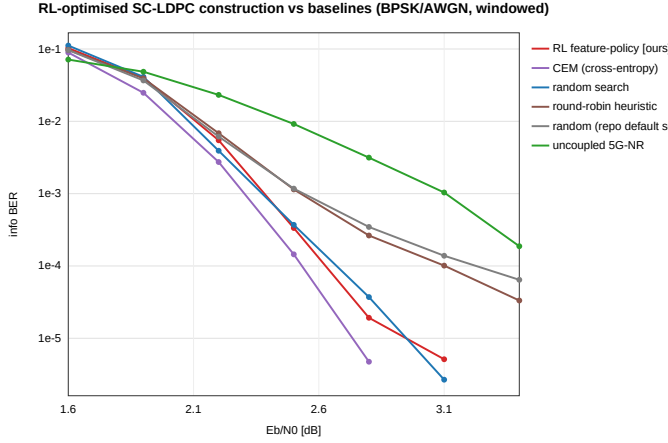


Fig. 1. BER-vs- E_b/N_0 waterfall for six constructions. Optimized constructions (RL/CEM/random search) decay without a floor; the repository default (seed-0) and the naive round-robin heuristic exhibit an error floor at $\approx 3\text{--}6 \times 10^{-5}$; the uncoupled component code is worst.

TABLE V
BEST VALIDATION FER VS. NUMBER OF EVALUATIONS (LEARNING CURVE).

Evaluations	~ 140	~ 260	~ 380	~ 500	~ 620
Random search	0.09	0.09	0.09	0.09	0.09
RL feature	0.09	0.09	0.075	0.065	0.060
RL per-edge	0.09	0.075	0.07	0.07	0.035
CEM	0.09	0.075	0.05	0.05	0.050

C. Sample Efficiency at Fixed Budget

Fig. 2 and Table V show the best validation FER versus number of evaluations. Random search stalls at ≈ 0.09 after about 140 evaluations (best-of- k is an order statistic with a very slow tail), while RL/CEM keep descending to 0.035–0.06 via a structured policy. On the 2000-frame retest, RL (0.080/0.092) and CEM (0.064) both consistently beat random search (0.133), and even the *deterministic greedy* rule (0.112)—which contains no sampling luck—beats random search.

D. Mechanism: Optimization Clears the 4-Cycles

All optimized constructions have $n_4 = 0$ and girth 6; the default/naive constructions have $n_4 \approx 930$ (round-robin 944, seed-0 928) and girth 4 (random reference: mean $n_4 = 525.6$, range $[0, 1408]$ over 40 samples). It is precisely these ≈ 930 4-cycles that cause the floor of the default/naive constructions; optimization clears the 4-cycles and the floor disappears. The learned 8-dimensional feature weights are listed in Table VI. Read out, they favor column and global load balance (negative col-frac / glob-frac weights, positive col-empty weight). We stress that this balance is a *soft heuristic, not the construction rule*: by the exact lemma of Section V, n_4 is set by (column-pair, shift-difference) coincidences within each check-block-row, which balance alone does *not* control—indeed a deterministic max-column-spread construction gives $n_4 = 960$, not 0, and the same-

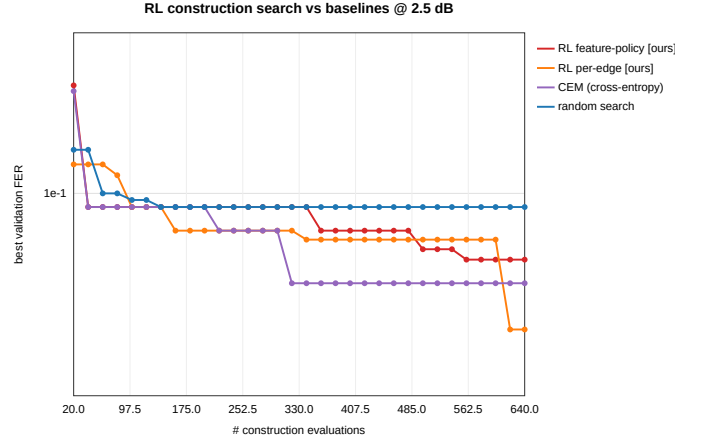


Fig. 2. Best validation FER vs. number of construction evaluations. Random search stalls at ≈ 0.09 ; RL (both policies) and CEM keep descending via a structured policy.

TABLE VI
LEARNED FEATURE-POLICY WEIGHTS (A SOFT HEURISTIC, NOT THE CONSTRUCTION RULE).

bias B_0	bias B_1	bias B_2	row-frac	col-frac	glob-frac	row-empty	col-empty
+0.33	+0.28	−0.12	−0.49	−1.85	−2.30	+0.69	+1.61

column/global co-assignment counts have ≈ 0 Spearman with n_4 . What actually drives $n_4 \rightarrow 0$ is the direct reduction of those coincidences: RL and CEM achieve it implicitly from the end-to-end FER reward, and a classical greedy/PEG/ACE local search (`classical_baseline.py`) achieves it explicitly from the exact 4-cycle count. The learned policy reaches the same 4-cycle-free outcome with no hand-coded cycle / trapping-set knowledge, but through a softer balance proxy rather than the closed-form rule that does not exist.

E. Full-Rate Sweep 0.5–0.9

We sweep the optimization across the rate range (low rate on BG2, high rate on BG1), auto-selecting the operating SNR per rate (typical random construction at $\text{FER} \approx 0.3$), with budget $64 \times 18 = 1152$ evaluations per optimizer. Table VII reports FER at the operating point and the waterfall-threshold gain at $\text{FER} = 0.1$ (seed-0 minus best optimizer). Three findings: (i) all optimizers beat the seed-0 default and the naive round-robin at *every* rate (0.005–0.39 vs. baseline 0.31–0.71), with a waterfall gain of 0.12–0.92 dB that is largest at moderate rate ($R \approx 0.6$, ~ 0.9 dB) and shrinks monotonically toward high rate (Fig. 3); this matches the threshold-saturation theory of Section IV. (ii) At this large budget the four optimizers are comparable—best-of-1152 random search is itself a strong baseline that erases RL’s sample-efficiency edge. (iii) The mechanism migrates with rate: at low/moderate rate the gain is driven by avoiding 4-cycles, while at high rate 4-cycles decouple from performance (e.g. a round-robin code with $n_4 = 3264$ is not bad at $R = 0.7$) and the dominant harmful structure is no longer the 4-cycle.

TABLE VII
FULL-RATE SWEEP: FER AT THE OPERATING POINT (1152
EVALS/OPTIMIZER) AND WATERFALL-THRESHOLD GAIN AT FER= 0.1.

R	BG	RL-f	RL-pe	CEM	rand	RR	seed0	gain
0.479	2	0.156	0.109	0.110	0.091	0.625	0.517	0.49 dB
0.603	2	0.098	0.112	0.098	0.095	0.644	0.709	0.92 dB
0.692	1	0.008	0.007	0.007	0.005	0.012	0.074	0.17 dB
0.770	1	0.360	0.385	0.279	0.386	0.499	0.696	0.14 dB
0.906	1	0.164	0.176	0.203	0.186	0.313	0.318	0.12 dB

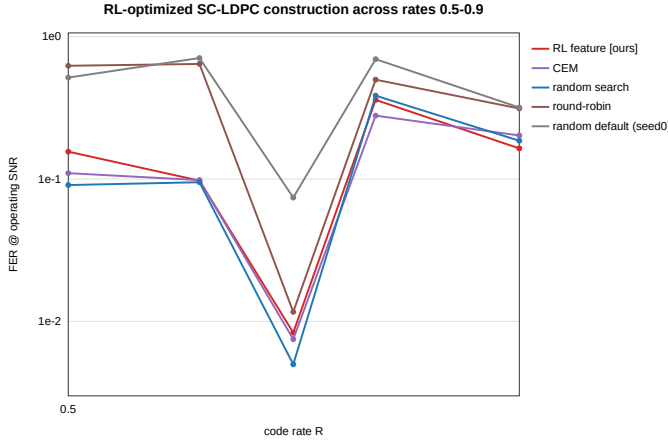


Fig. 3. Waterfall-threshold construction gain (seed-0 minus best optimizer at FER= 0.1) vs. code rate. The gain is largest at moderate rate ($R \approx 0.6$) and shrinks toward high rate, consistent with threshold saturation closing the BP-MAP gap (Section IV).

F. The RL Home Court: Long Codes and Large w

We now test the central thesis—RL’s advantage appears when evaluation is *expensive* and the space is *large*. We move to long 5G-NR BG1 codes ($Z=32$, component codewords ≈ 860 –1470 bits), high wireless rates $\{1/2, 2/3, 3/4, 5/6, 7/8\}$, and coupling memory $w \in \{1, 2, 3\}$, on 2×80 core cluster pods. A single frame costs ~ 0.9 s, so each method gets only a small budget of ~ 640 evaluations—sample efficiency becomes the hard currency—and the search space $(w+1)^E$ ($E=69$ –172) reaches 4^{172} at $w=3$, far beyond what random search can cover. Table VIII gives FER at the operating point.

Findings. (i) RL (feature or per-edge) beats same-budget random search on 13/15 cells—in sharp contrast to the “tie at large budget” of the mid-scale sweep. (ii) The advantage grows with w : $w=1 \rightarrow 4/5$, $w=2 \rightarrow 4/5$, $w=3 \rightarrow 5/5$, and RL often drives FER to exactly 0 while random search stalls at 0.05–0.14 (e.g. $R=2/3, w=3$: RL 0.000 vs. random 0.140). (iii) RL dominates the repository default across the board. (iv) Honestly, the two exceptions ($R=5/6, w=1$ and $R=7/8, w=2$) both occur at small w where the space is small and random search suffices, and the $R=1/2, w=1$ auto-operating point is low (FER ≈ 0.9 , poor discrimination). Fig. 4 plots the RL/random advantage by rate and w .

G. Pushing w to 12: Advantage Grows Monotonically

We extend the expensive Monte-Carlo evaluation to $w \in \{3, 5, 8, 12\}$ (window $W = w+1$), full rate, 20 cells, on

TABLE VIII
LONG-CODE, HIGH-RATE, LARGE- w CELLS (BG1, $Z=32$): FER AT THE
OPERATING POINT.

R	w	E	SNR	RL-f	RL-pe	CEM	rand	seed0
1/2	1	172	1.5	0.913	0.900	0.910	0.907	0.993
1/2	2	172	2.0	0.017	0.043	0.073	0.083	0.707
1/2	3	172	2.5	0.010	0.010	0.080	0.063	0.830
2/3	1	121	2.5	0.040	0.057	0.083	0.080	0.127
2/3	2	121	2.5	0.090	0.117	0.100	0.197	0.917
2/3	3	121	3.0	0.000	0.010	0.017	0.140	0.773
3/4	1	96	3.0	0.033	0.060	0.033	0.047	0.077
3/4	2	96	3.0	0.110	0.120	0.027	0.120	0.670
3/4	3	96	3.5	0.003	0.000	0.040	0.020	0.527
5/6	1	75	3.5	0.200	0.240	0.180	0.167	0.430
5/6	2	75	3.5	0.293	0.320	0.313	0.330	0.737
5/6	3	75	4.0	0.010	0.037	0.067	0.027	0.400
7/8	1	69	4.0	0.040	0.067	0.077	0.080	0.097
7/8	2	69	4.0	0.117	0.123	0.147	0.073	0.290
7/8	3	69	4.5	0.000	0.007	0.057	0.050	0.280

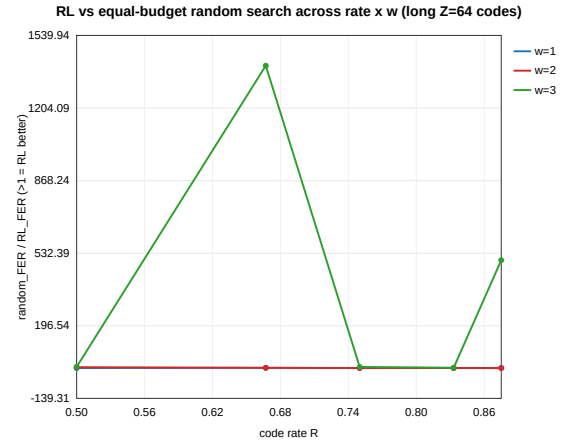


Fig. 4. RL-vs-random-search advantage across the long-code cells, by rate and coupling memory w . The advantage is systematic and grows with w (search-space size).

an eviction-resistant 100-core pod. Table IX shows the trend cleanly: as w grows from 3 to 8, random search gets steadily worse (e.g. $R=2/3$: 0.040 $\rightarrow$ 0.083 $\rightarrow$ 0.133) while RL gets steadily better, often reaching exactly 0 (0.017 $\rightarrow$ 0.010 $\rightarrow$ **0.000**). RL’s win rate by w is $w=3 : 5/5$, $w=5 : 3/4$, $w=8 : 4/4$, $w=12 : 2/2$ (over cells with signal; the rest are ties, none losses). The default construction is almost uniformly 0.7–1.0 (fully broken). Four “no-signal” cells (where all methods give FER= 1.0) arise because the large- w rate loss pulls the SC rate too low (e.g. $R=1/2, w=12 \Rightarrow$ SC 0.427) so the auto operating point falls below the waterfall—not a decoding bug; the same $w=8$ is fine at $R=2/3$.

H. Cheap Surrogate at $w \leq 30$: a Clear “When to Use RL” Boundary

End-to-end Monte-Carlo is infeasible at very large w (the window must satisfy $W \geq w+1$; ~ 18 s/frame at $w=30, L=300$), so we switch the reward to a fast structural surrogate—minimize the lifted-graph 4-cycle count (~ 0.3 s, independent of w)—and sweep $w \in \{3, 5, 8, 12, 16, 20, 30\}$, $L \in \{30, \dots, 240\}$. Table X reports n_4 at $L=60$. Both

TABLE IX
WIDE-MEMORY MONTE-CARLO SWEEP (RL / RANDOM-SEARCH FER).
“—” DENOTES A NO-SIGNAL CELL (ALL METHODS FER= 1.0).

R	$w=3$	$w=5$	$w=8$	$w=12$
1/2	0.000 / 0.037	—	—	—
2/3	0.017 / 0.040	0.010 / 0.083	0.000 / 0.133	—
3/4	0.007 / 0.033	0.010 / 0.010	0.000 / 0.010	0.003 / 0.087
5/6	0.007 / 0.063	0.007 / 0.143	0.000 / 0.203	—
7/8	0.003 / 0.100	0.003 / 0.023	0.000 / 0.043	0.010 / 0.037

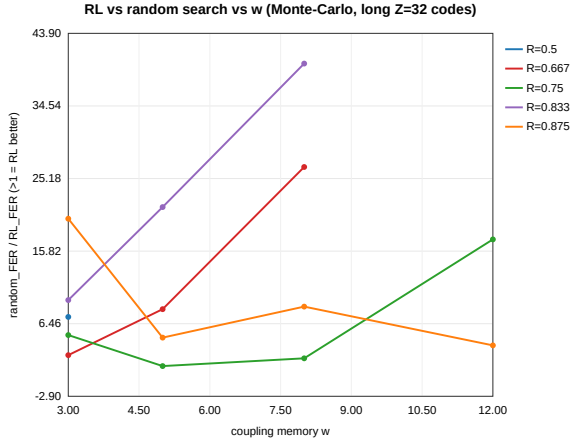


Fig. 5. Wide-memory ($w \leq 12$) Monte-Carlo sweep: RL vs. random-search FER. As w grows, random search degrades while RL improves toward 0, nailing the “advantage grows with evaluation cost $\times$ space” rule.

optimization (RL and random search) drive n_4 to 0 at every w , while the single seed-0 construction is unreliable (≈ 900 at $w=5-16$). Under this *cheap* surrogate, $RL \approx$ random search—which is exactly the point: when evaluation is cheap, a brute-force best-of-1400 reaches the same $n_4=0$, and RL has no extra advantage. RL pays off only when the objective is *expensive* (long-code Monte-Carlo, density evolution / PEXIT). This unifies the three regimes into one rule: *RL’s value rises monotonically with per-evaluation cost*.

I. Zero-Shot Transfer

We move the 8 learned feature weights *unchanged* to two new codes and greedily construct (no search on the new code). Table XI shows the zero-shot rule beats both the typical random construction (median) and the round-robin heuristic on both new configurations, indicating that RL learned a genuine structural principle rather than overfitting one code. This is a deployment property random search / CEM cannot provide, since they output a code-specific assignment that must be re-researched for every new code. (A best-of-12 random search can occasionally edge out the single zero-shot greedy code—an unequal 12-vs-0-evaluation comparison.)

J. Threshold (PEXIT) Objective Results

The PEXIT objective of Section IV is specified and integrated into the pipeline, but its numerical validation is still pending; see Section IV-D and the placeholder Table II.

TABLE X
CHEAP 4-CYCLE SURROGATE, $w \leq 30$: n_4 AT $L=60$ (LOWER IS BETTER).

w	3	5	8	12	16	20	30
RL feature	0	0	0	0	0	0	0
RL per-edge	0	0	0	0	0	0	0
Random search	0	0	0	0	0	0	0
Default seed-0	0	960	912	928	928	0	0

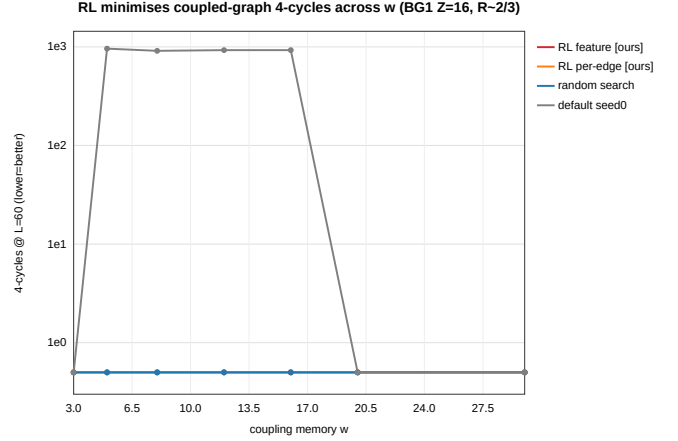


Fig. 6. n_4 at $L=60$ vs. coupling memory w under the cheap 4-cycle surrogate. Optimization (RL or random) reaches $n_4=0$ for all w ; the single seed-0 construction is unreliable. Under a cheap objective, $RL \approx$ random search.

[TODO: insert PEXIT numerical results (Spearman, PEXIT-reward champion MC re-test, eval-cost ratio) once the experiment of Section IV-D is run.]

VIII. DISCUSSION AND LIMITATIONS

A. What RL Buys, Honestly

The hardest, most robust conclusion is construction-level, not RL-level: optimizing edge spreading lowers FER by 5–11 $\times$ and eliminates the error floor, and the repository default is among the worst constructions. RL’s specific value is threefold: it *drives the 4-cycle count to zero* from the FER reward alone—without any closed-form rule, since none exists (Section V)—it is *more sample-efficient* than same-budget random search, and its feature policy *transfers zero-shot*.

B. Conditional, Not Unconditional, Wins

RL’s advantage over the baselines is *conditional*, and we present this as an honest feature rather than a weakness. At a large evaluation budget, best-of- N random search is a strong baseline and CEM ties or slightly beats policy-gradient RL on single-point FER (e.g. CEM 0.064 vs. RL feature 0.092 in Table III); in the deep waterfall, several 4-cycle-free constructions converge within Monte-Carlo noise. The first-order factor is whether a construction avoids 4-cycles, which several optimizers can achieve; RL’s incremental value is automation, interpretability, and transfer—and a clean criterion for *when* sample efficiency matters (Sections VII-F–VII-H).

TABLE XI
ZERO-SHOT TRANSFER OF THE 8 LEARNED WEIGHTS (GREEDY, NO RE-SEARCH): FER ON NEW CODES.

Transfer target	rate	zero-shot	random (med.)	round-robin
$Z=24$ (longer blocks)	0.603	0.124	0.146	0.240
$m_p=6$ (higher rate)	0.693	0.788	0.852	0.960



Fig. 7. Zero-shot transfer of the learned feature policy to new codes. The transferred rule (greedy, no re-search) is below the random-construction median and the round-robin heuristic on both new configurations.

C. Scale and Scope

This is an empirical finite-length study with no new asymptotic theorem: the reward is single-operating-point Monte-Carlo FER rather than a closed-form or learned objective (the PEXIT objective of Section IV is the principled remedy, with numerics pending). Experiments are mid-scale and CPU-only ($Z \leq 128$, $w \leq 30$, $L \leq 300$); larger lifting and hardware are not a contribution dimension here—indeed, the “expensive evaluation” argument is exactly why the criterion holds even at mid scale. Large- w high-rate Monte-Carlo has a calibration difficulty: coupling rate loss steepens strong-code waterfalls, so a few auto operating points fall off the waterfall (FER=1.0), which a more robust operating-point selection or a threshold-type objective would fix.

D. A Scoped Negative Result on the Error Floor

We tested whether a cheap absorbing-set surrogate could *directly* target the finite-length error floor, and the answer is negative—an honest, scoped result that points to future work. In an overnight campaign, the measured error floor spans about 2.5 orders of magnitude across constructions (so the floor is strongly construction-dependent), yet the Spearman correlation between the cheap absorbing-set count ($a \leq 6$) and the floor is only $\rho \approx -0.13$; minimizing the absorbing-set count (cem_floor) gave a *worse* floor (6.1×10^{-7}) than the waterfall-objective optimizer (cem_waterfall, 3.4×10^{-8}). Moreover, the feature policy that transferred zero-shot under the waterfall objective *degenerated* under the floor objective (collapsing to an all-zero assignment with a much larger absorbing-set count and a floor of its own). The takeaway:

the standard cheap absorbing-set enumeration does *not* predict this class of finite-length floors, and minimizing it can even be harmful. A faithful floor objective therefore needs either expensive Monte-Carlo or a *learned* neural floor surrogate that amortizes the otherwise-infeasible trapping-set enumeration—the natural next direction (and outside the scope of this paper).

E. Future Work

In leverage order: (1) replace/augment the reward with a DE/PEXIT threshold (the sharpest instance of the “expensive evaluation $\Rightarrow$ RL wins” thesis); (2) complete the numerical lemma validation (Section V-E); (3) jointly optimize the QC lifting shifts (further enlarging the space and RL’s home court) and meta-learn across rates/SNRs to produce code families; (4) combine orthogonally with RL decoding—learn how to *build* and how to *decode* together; and (5) the learned neural floor surrogate of Section VIII-D.

IX. CONCLUSION

Edge spreading is a high-leverage but routinely randomized construction freedom in 5G-NR-based SC-LDPC codes. Casting it as an MDP and optimizing it end-to-end with policy gradient—using the real-chain FER as reward—lowers FER by 5–11 $\times$, eliminates the error floor across rates 0.5–0.9 (waterfall gain 0.12–0.92 dB, largest at $R \approx 0.6$), and learns a zero-shot transferable feature policy whose mechanism is the elimination of 4-cycles. We characterize that mechanism with an exact, bit-exact-validated combinatorial lemma, $n_4 = \sum_{\text{keys}} Z \binom{m}{2}$ over (check-block-row, column-pair, shift-difference) keys, which shows why no simple balance rule removes the 4-cycles (the shift-differences are fixed by the 5G-NR circulants) and hence why search-based construction is needed. Beyond the codes themselves, the work contributes a deployable methodological criterion: the sample-efficiency advantage of reinforcement learning over blind search rises monotonically with per-evaluation cost and search-space size—RL is not universally superior, but in the expensive-evaluation, large-space regime of real finite-length code design it is systematically better than blind search.

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